Seminar 1

Preferences and utility

Before the exercises

These two topics were left for the seminar. The teaching assistant introduces the definitions and examples below before moving to the problem sheet.

Separate drawers, one wardrobe: separability

Write a bundle as (y, z), with $y = (r, s)$ describing a computer's processing power and storage, and $z \geq 0$ the quality of the workspace. The computer group is *separable* from the workspace if, for every y, y', z, z',

$$(y, z) \succeq (y', z) \quad \Longleftrightarrow \quad (y, z') \succeq (y', z').$$

The ranking of two computers stays the same when the common workspace changes. One way to model this is $u(y, z) = U(v(y), z)$, with U strictly increasing in the group index v. For example,

$$v(r, s) = \log(1 + r) + 2\log(1 + s), \qquad U(v, z) = (1 + v)(1 + z).$$

This representation is not additively separable. Separability does not fix the amount spent on the computer: it fixes a within-group ranking.

The same price tag in every wallet: quasilinearity

Let $a \geq 0$ be money left after other purchases and $q \in [0, 3]$ a concert experience. A utility function quasilinear in a has the form

$$u(a, q) = a + h(q), \qquad \text{for example } h(q) = 6q - q^2.$$

Adding the same feasible amount to both wallets preserves their comparison:

$$(a, q) \succeq (a', q') \quad \Longleftrightarrow \quad (a + t, q) \succeq (a' + t, q').$$

Compensation between two concert experiences does not depend on starting wealth. A transformed index such as $e^{a+h(q)}$ need not itself have the displayed linear form, although its preferences admit a quasilinear representation. With $a \geq 0$, low-income corners can still create income effects.

Exercises

Give a short justification for each answer. Specify the domain and bundles in a counterexample. Quantities are nonnegative unless stated otherwise. All continuity and neighbourhood claims use the stated domain's relative Euclidean topology.

1. Five stars, rounded down

A platform assigns $u(x) = \lfloor x_1 \rfloor + \lfloor x_2 \rfloor$ on $\mathbb{R}^2_+$, where $\lfloor t \rfloor$ is the greatest integer not exceeding t. Are the represented preferences rational? Are they continuous? For the latter, construct a sequence of comparisons, rather than merely pointing to a discontinuous formula.

2. The brighter of two spotlights

For $u(x) = \max\{x_1, x_2\}$, sketch the indifference set at utility 2 and its weak upper contour set. Are preferences convex, monotone, strongly monotone and locally nonsatiated? Monotonicity here means that increasing *every* good strictly improves the bundle.

3. A plan for rain and shine

A holiday bundle $x = (h, b, m) \in \mathbb{R}^3_+$ includes hotel quality, beach access and museum access. Its scores are $S(x) = h + 2b$ in sunshine and $R(x) = h + 3m$ in rain. Define $x \succeq y$ iff $S(x) \geq S(y)$ *and* $R(x) \geq R(y)$. Is the relation complete? Transitive? Continuous? Can one real-valued utility represent it? Distinguish "neither plan is worse in every scenario" from indifference.

4. An extra drop spoils the deal

Let $u(x) = x_1 - \mathbf{1}\{x_2 > 0\}$, with **1** equal to 1 when its condition holds and 0 otherwise. Is the consumer locally nonsatiated? Are preferences continuous at the boundary $x_2 = 0$? Can local nonsatiation coexist with a discontinuity?

5. A ceiling you can never touch

On $\mathbb{R}^2_+$, $u(x) = \arctan(x_1 + x_2)$. Is utility bounded above? Is there a best bundle on the entire domain? Are preferences locally nonsatiated? What does this say about inferring satiation from a numerical utility bound?

6. Keep the seesaw level

Let $u(x) = -|x_1 - x_2|$. Describe all weak upper contour sets. Are preferences convex? Strictly convex? Find *all* bundles at which local nonsatiation fails.

7. The shop skips the small sizes

A product is available only in sizes $X = \{0\} \cup [1, \infty)$, and $x \succeq y$ iff $x \geq y$. Are preferences continuous and strongly monotone on this domain? Are they locally nonsatiated at 0? Does this contradict any implication stated in the lecture on $\mathbb{R}^L_+$?

8. A wristband beats the whole queue

A festival ticket is $(d, q) \in \{0, 1\} \times [0, 1]$: d indicates VIP status and q seat quality. Every VIP ticket beats every regular ticket; within a class, higher q wins. Construct a continuous real-valued utility representation. Can you still do so if $q \in \mathbb{R}_+$ is unbounded? Explain why the square-domain lexicographic impossibility result does not decide this problem.

9. Two batteries, both fully charged

For $u(x) = \min\{x_1, 1\} + \min\{x_2, 1\}$, draw the indifference set at utility 1. Are preferences convex? Find all bundles at which local nonsatiation fails. Is the failure confined to one bliss point?

10. The interview panel's missing shortlist

An employer reports the *full* sets of best candidates:

$$C(\{a, b\}) = \{a, b\}, \qquad C(\{b, c\}) = \{b, c\}, \qquad C(\{a, c\}) = \{a\}.$$

Can one rational ranking generate these reports? If the reports instead list only candidates actually invited, could an uninvited candidate still be tied for best? Give one ranking compatible with that weaker reading of all three reports.

11. Two knobs, one wallet

A podcaster chooses microphone quality r, lighting quality s and money left for everything else a. Define the studio index and overall utility by

$$v(r,s) = \sqrt{1+r} + \sqrt{1+s} - 2, \qquad u(a,r,s) = a + 6v(r,s).$$

All quantities are nonnegative; all prices are 1; the budget is $a+r+s \le m$, where $m > 0$.

(a) Is the studio group separable from a? Is the utility quasilinear in a? Explain the economic meaning of each property here.

(b) Show that any exact index $v \ge 0$ is cheapest to achieve with equal upgrades

$$r = s = v + \frac{v^2}{4}, \qquad \text{at cost } c(v) = 2v + \frac{v^2}{2}.$$

Hint: set $R = \sqrt{1+r}$, $S = \sqrt{1+s}$ and use $R^2 + S^2 \ge (R+S)^2/2$. Include $v = 0$.

(c) Explain why an optimum exhausts the budget and uses the cheapest studio bundle for its index. Reduce the problem to choosing v and state its feasible interval.

(d) Find the optimal (a,r,s) at $m = 6$, $m = 16$ and $m = 30$, then for general $m > 0$. No Lagrangian is needed: complete a square in the reduced objective.

(e) As income rises, when do the studio upgrades stop changing? Does this mean separability fixes studio spending at every income? Locate the income range in which the nonnegativity of a matters.